KO LIH HAN

Software Engineer | Applied AI & LLM Systems

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EXPERIENCE

Software Developer (Part-time)

Kahap Inc — Aug 2022 – Aug 2026

- Led 0-to-1 delivery of a production feedback platform, owning backend workflows, deployment, administration, and ongoing maintenance.
- Centralized submissions, discussions, notifications, and admin workflows into one production system, replacing fragmented manual handling.
- Maintained a Laravel/Vue.js career platform across queues, scheduled jobs, database issues, SSL renewals, and Linux server migrations.
- Integrated GPT-SoVITS API workflows and automated text/pronunciation checks into production features.

Full-Stack Developer Intern

Pac Com Solution — Jun 2021 – Jan 2022

- Developed and maintained Laravel/Vue.js features and frontend/backend fixes for a production POS system.

PROJECTS

Small-Agent QLoRA — 2026

Python, Qwen3.5-9B, QLoRA / PEFT, Transformers, llama.cpp

- Fine-tuned a Qwen3.5-9B tool agent with QLoRA, raising exact task success from 34.5% to 62.0% across a frozen 200-task evaluation.
- Built a bounded agent runtime with search, file, and Python tools plus duplicate-call blocking, timeouts, and trace logging.
- Validated the tuned model on GAIA100, improving accuracy from 9% to 18% over the base model; trace analysis identified over-search as the main remaining failure mode.

Semantic Text-to-SQL — 2026

Python, FastAPI, LangGraph, SQLite, Qwen3.5-9B

- Built a local Text-to-SQL service with schema grounding, deterministic validation, bounded repair, and read-only SQLite execution.
- On 100 frozen BIRD DEV queries, improved conservative execution success from 75% to 90% and execution-match accuracy from 33% to 39%, with 6 wrong-to-correct recoveries and 0 regressions.
- Exposed the same service through FastAPI, CLI, and Streamlit interfaces while keeping verification and evaluation logic in one backend layer.

RESEARCH

Context-Aware Workplace Language Understanding — 2025 – 2026

M.S. Thesis, National Tsing Hua University

- Built a labeled workplace-dialogue dataset with 20,455 utterances across 7 boards for implicit intent and listener-side interpretation.
- Combined role and context graphs with retrieval, improving target-only macro-F1 from 0.625 to 0.689 on 401 test utterances.

SKILLS

AI / LLM: Hugging Face Transformers, PEFT / QLoRA, LangGraph, llama.cpp, RAG, tool calling, LLM and retrieval evaluation

Backend: Python, FastAPI, SQL, PHP, Laravel, JavaScript, REST APIs

Data / Tools: MySQL, SQLite, Git, GitHub Actions, pytest

Frontend: Vue.js, React

Languages: Chinese (Native/Bilingual), English (Fluent), Malay (Intermediate)

EDUCATION

National Tsing Hua University — Jun 2026

Master of Science, Information Systems & Applications | GPA: 4.09/4.3

Chung Hua University — Jun 2024

Bachelor of Science, Computer Science & Information Engineering | GPA: 3.9/4.0